www.RocketRevise.com

Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Isotopes are atoms of an element that have different

- ☐ **A** electronic structures
- ☐ **B** numbers of electrons
- ☐ **C** numbers of protons
- ☐ **D** numbers of neutrons

RocketRevise

(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the electronic configuration of the Sc^{3+} ion?

- ☐ **A** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$
- ☐ **B** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5\ 3d^1$
- ☐ **C** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1\ 4s^2$
- ☐ **D** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^4\ 4s^2$

(Total for question = 1 mark)

Q3.

RocketRevise

Which element is in the d-block of the Periodic Table?

(1)

- ☐ **A** argon
- ☐ **B** chlorine
- ☐ **C** iron
- ☐ **D** sodium

(Total for question = 1 mark)

www.RocketRevise.com

www.RocketRevise.com

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the equation for the **second** ionisation energy of an element, A?

- ☒ **A** $A(g) \rightarrow A^{2+}(g) + 2e^-$
- ☒ **B** $A^+(g) \rightarrow A^{2+}(g) + e^-$
- ☒ **C** $A^{2+}(g) \rightarrow A^{3+}(g) + e^-$
- ☒ **D** $A^{2+}(g) \rightarrow A^{4+}(g) + 2e^-$

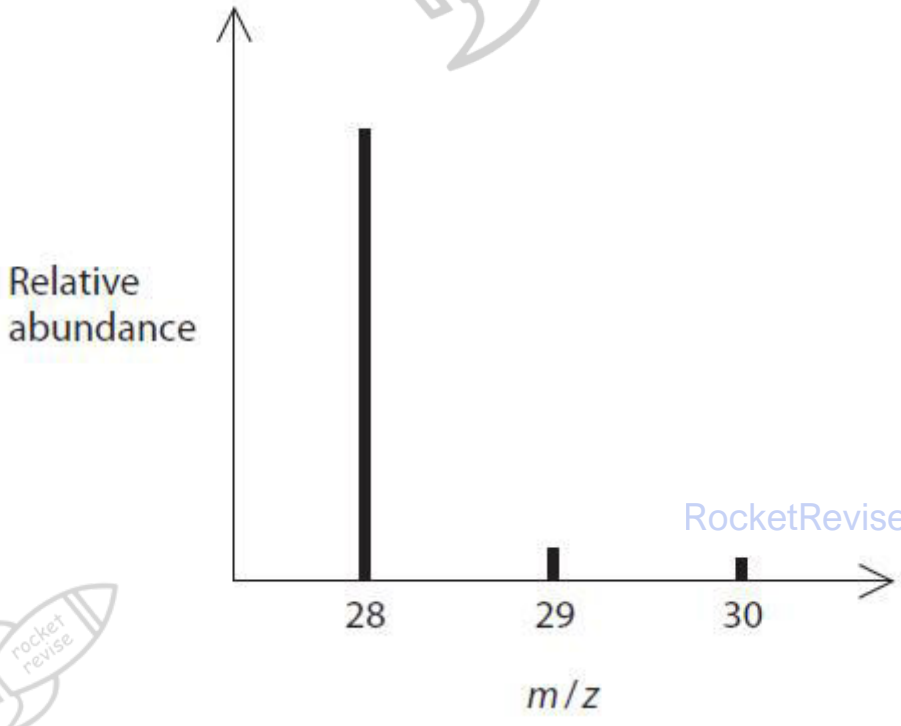
(Total for question = 1 mark)

RocketRevise

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The mass spectrum of a sample of silicon is shown.



RocketRevise

What is the **best** estimate for the relative atomic mass of silicon in this sample?

- ☒ **A** 28.0
- ☒ **B** 28.2
- ☒ **C** 28.8
- ☒ **D** 29.0

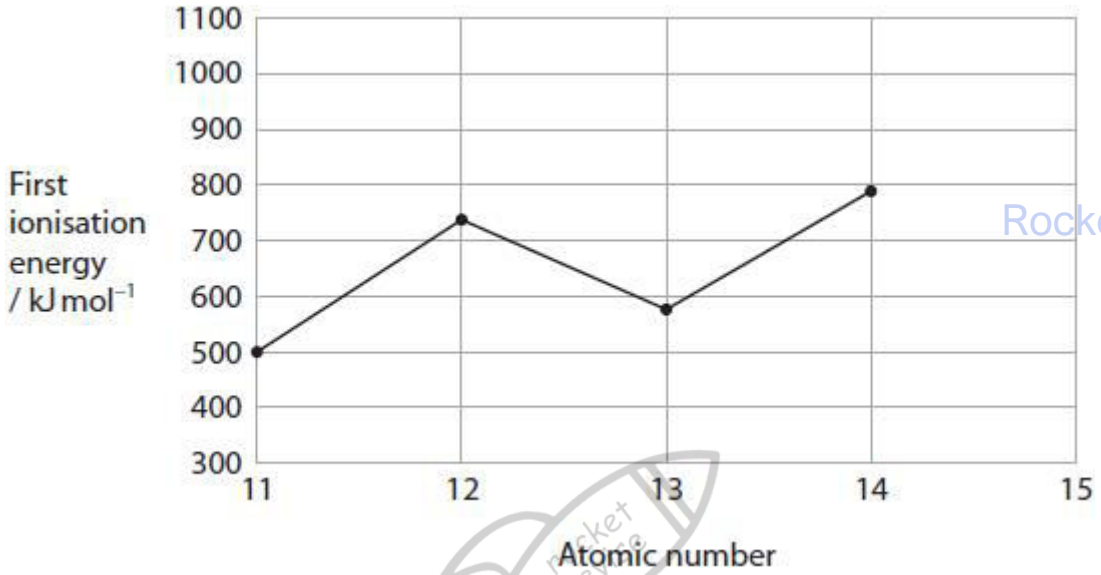
www.RocketRevise.com

www.RocketRevise.com

(Total for question = 1 mark)

Q6.

The diagram shows the first ionisation energy for the elements from sodium to silicon.



What is the approximate first ionisation energy, in kJ mol^{-1}, of phosphorus (atomic number 15)?

- ☒ **A** 400
- ☒ **B** 500
- ☒ **C** 700
- ☒ **D** 1000

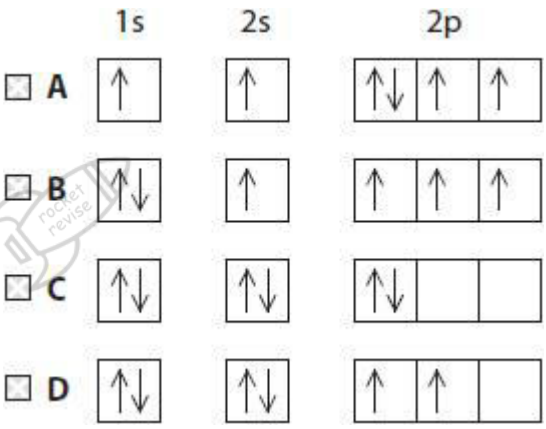
(Total for question = 1 mark)

Q7.

Which is the electronic configuration of a carbon atom in its ground state?

www.RocketRevise.com

www.RocketRevise.com



(Total for question = 1 mark)

RocketRevise

Q8.

The first six ionisation energies of an element, in kJ mol^{-1}, are shown.

578 1817 2745 11578 14831 18378

Which group of the Periodic Table includes this element?

- **A** Group 2
- **B** Group 3
- **C** Group 4
- **D** Group 5

(Total for question = 1 mark)

RocketRevise

Q9.

What is the maximum number of electrons in the 3p subshell, and in the third quantum shell of an atom?

www.RocketRevise.com

www.RocketRevise.com

	Maximum number of electrons in the 3p subshell	Maximum number of electrons in the third quantum shell
☐ A	2	8
☐ B	2	18
☐ C	6	8
☐ D	6	18

(Total for question = 1 mark)

RocketRevise

Q10.

The hydroxide ion, OH^-, has a total of 9 protons.

How many neutrons and electrons are there in this ion?

	Number of neutrons	Number of electrons
☐ A	8	8
☐ B	8	10
☐ C	9	8
☐ D	9	9

(Total for question = 1 mark)

Q11.

RocketRevise

A sample of silicon contains the following isotopes.

Isotope	Percentage abundance
^{28}Si	81.21
^{29}Si	14.10
^{30}Si	4.69

What is the relative atomic mass of silicon, to one decimal place, in this sample?

☐ **A** 28.0

www.RocketRevise.com

www.RocketRevise.com

- **B** 28.2
- **C** 29.0
- **D** 29.8

(Total for question = 1 mark)

Q12.

The decrease in first ionisation energy down Group 1 in the Periodic Table is caused by the **increase** in

(1)

- **A** force of attraction between the nucleus and outer electron
- **B** number of neutrons in the nucleus
- **C** number of protons in the nucleus
- **D** shielding of the outer electron from the nuclear charge

(Total for question = 1 mark)

Q13.

What is the equation for the **third** ionisation energy of aluminium?

(1)

- **A** $Al(g) \rightarrow Al^{3+}(g) + 3e^-$
- **B** $Al(s) \rightarrow Al^{3+}(s) + 3e^-$
- **C** $Al^{2+}(g) \rightarrow Al^{3+}(g) + e^-$
- **D** $Al^{2+}(s) \rightarrow Al^{3+}(s) + e^-$

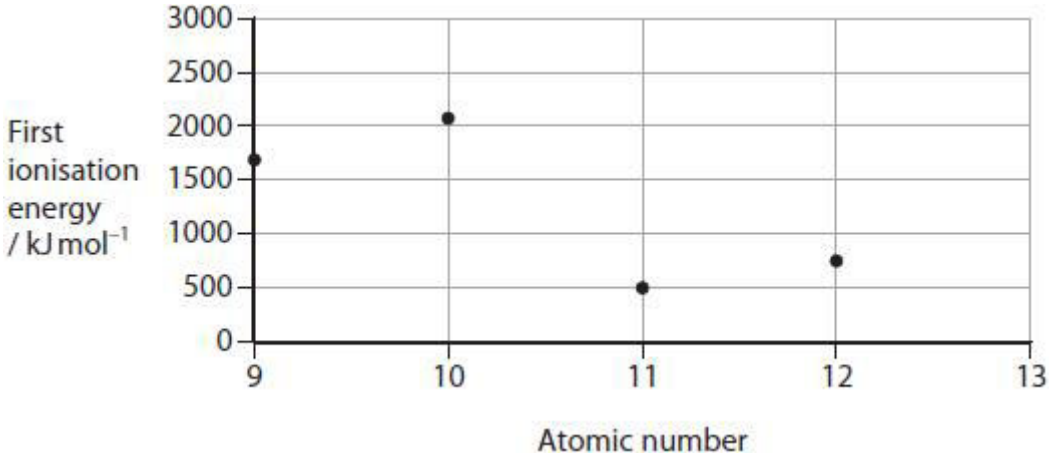
(Total for question = 1 mark)

Q14.

The chart shows the first ionisation energy of each of the elements from fluorine to magnesium.

www.RocketRevise.com

www.RocketRevise.com



Which is the approximate first ionisation energy, in kJ mol^{-1}, of aluminium (atomic number 13)?

RocketRevise

(1)

- ☒ **A** 300
- ☒ **B** 600
- ☒ **C** 900
- ☒ **D** 1200

(Total for question = 1 mark)

Q15.

The first three ionisation energies of carbon are shown.

	1st	2nd	3rd
Ionisation energy / kJ mol^{-1}	1086	2353	4621

Which is the approximate fourth ionisation energy, in kJ mol^{-1}, of carbon?

(1)

- ☒ **A** 3500
- ☒ **B** 6200
- ☒ **C** 11 000
- ☒ **D** 38 000

RocketRevise

(Total for question = 1 mark)

Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

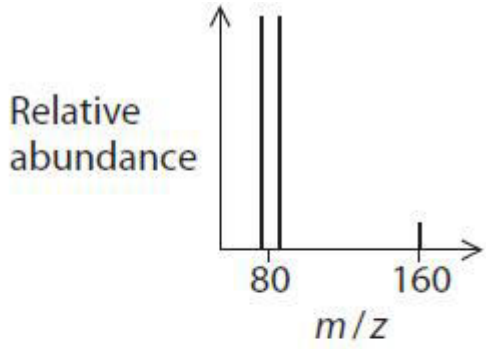
www.RocketRevise.com

www.RocketRevise.com

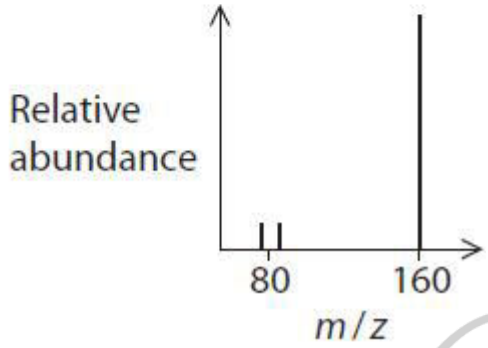
The two stable isotopes of bromine have relative masses of 79 and 81.

Which is the correct pattern of peaks in the mass spectrum of molecular bromine?

☒ **A**

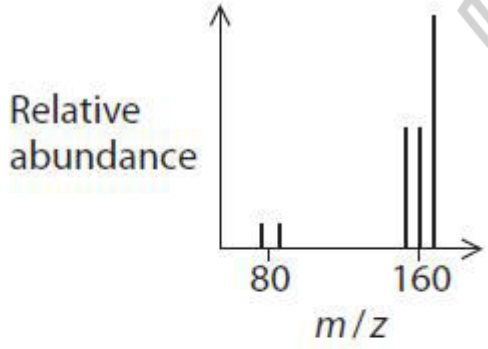


☒ **B**

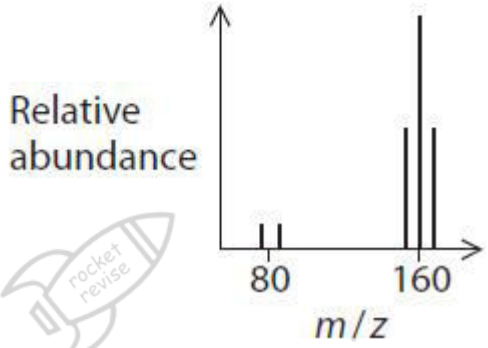


RocketRevise

☒ **C**



☒ **D**



RocketRevise

(Total for question = 1 mark)

Q17.

www.RocketRevise.com

www.RocketRevise.com

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which statement about subatomic particles is correct?

- ☒ **A** neutral atoms always contain the same number of protons and electrons
- ☒ **B** neutral atoms always contain the same number of protons and neutrons
- ☒ **C** electrons have a relative mass of 1 and a charge of −1
- ☒ **D** protons have a relative mass of 1 and no charge

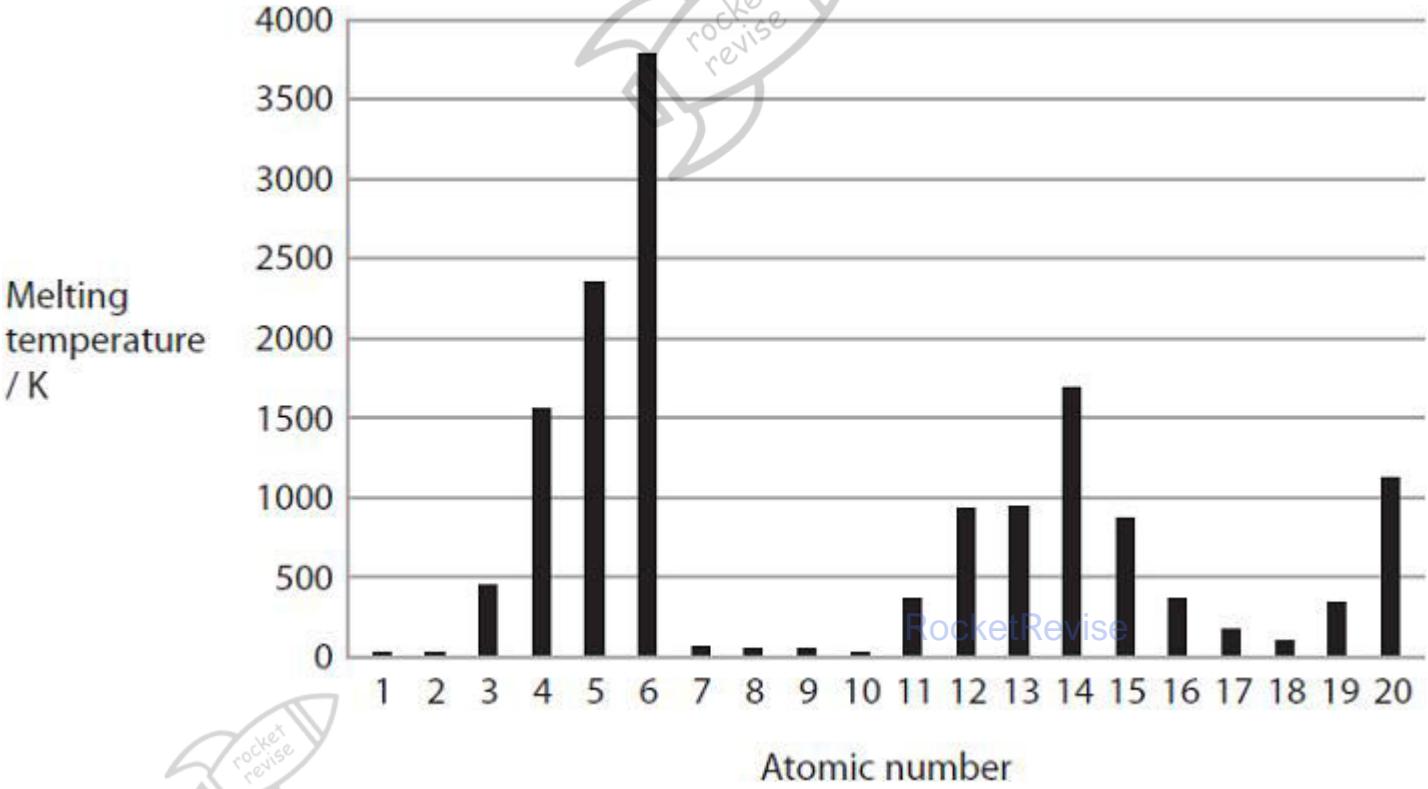
(Total for question = 1 mark)

RocketRevise

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The bar chart shows the melting temperatures of the first twenty elements.



The bar chart shows that melting temperatures

- ☒ **A** of giant covalent structures are the highest in Period 2 and in Period 3
- ☒ **B** of metals are always higher than non-metals
- ☒ **C** increase going down each group
- ☒ **D** increase across Period 2 and Period 3

(Total for question = 1 mark)

www.RocketRevise.com

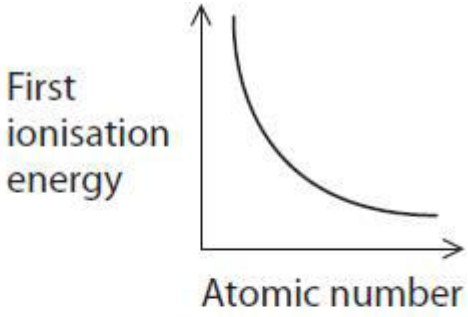
www.RocketRevise.com

Q19.

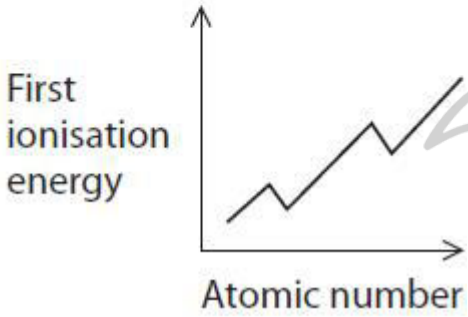
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which sketch graph shows the trend in first ionisation energy values going down Group 1 in the Periodic Table?

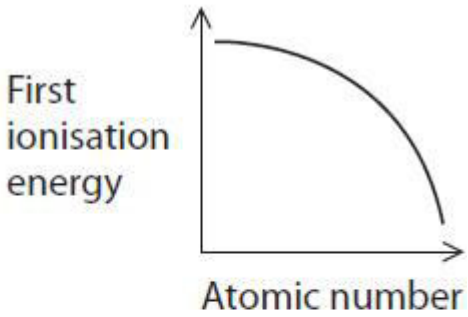
☐ **A**



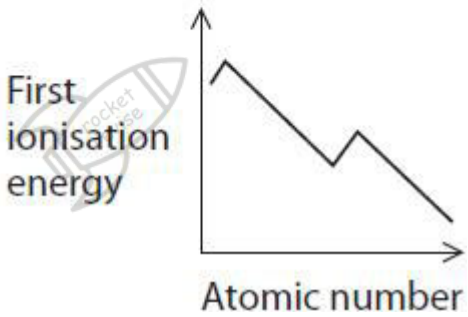
☐ **B**



☐ **C**



☐ **D**



(Total for question = 1 mark)

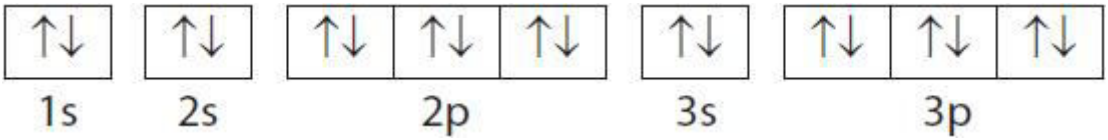
www.RocketRevise.com

www.RocketRevise.com

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which ion does **not** have the electronic configuration shown?



- ☐ **A** K^+
- ☐ **B** Ca^{2+}
- ☐ **C** Ti^{2+}
- ☐ **D** Sc^{3+}

RocketRevise

(Total for question = 1 mark)

Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the electronic configuration of the nitride ion, N^{3-} ?

- ☐ **A** $1s^2\ 2s^2$
- ☐ **B** $1s^2\ 2s^2\ 2p^3$
- ☐ **C** $1s^2\ 2s^2\ 2p^4$
- ☐ **D** $1s^2\ 2s^2\ 2p^6$

(Total for question = 1 mark)

RocketRevise

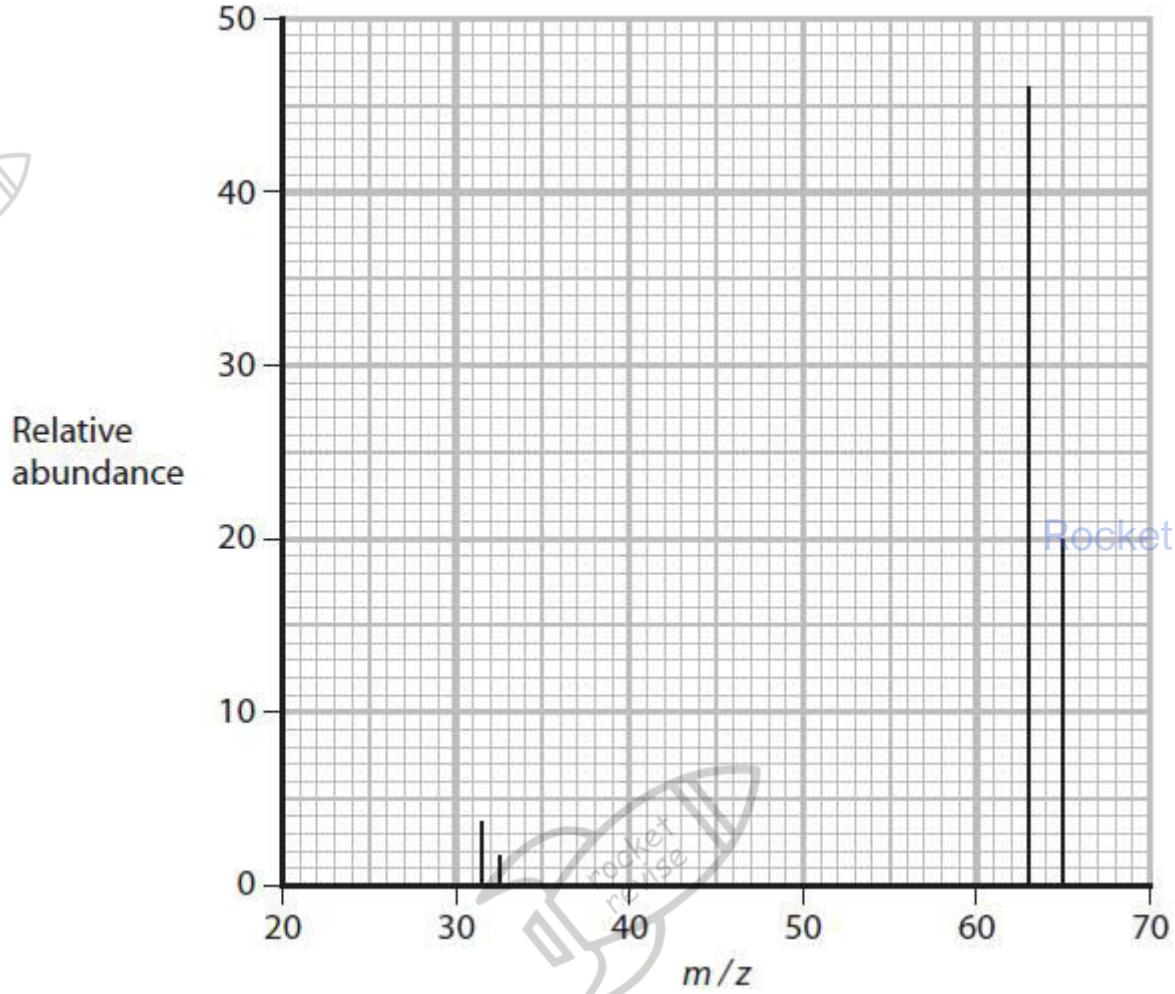
Q22.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The mass spectrum of a sample of an element is shown.

www.RocketRevise.com

www.RocketRevise.com



(a) What is the A_r of the element?

(1)

- ☒ **A** 42.0
- ☒ **B** 48.0
- ☒ **C** 63.6
- ☒ **D** 64.0

(b) Which species could be responsible for the peak at $m / z = 32.5$?

(1)

- ☒ **A** $^{32}S^{+}$
- ☒ **B** $^{63}Cu^{2+}$
- ☒ **C** $^{65}Cu^{2+}$
- ☒ **D** $^{98}Tc^{3+}$

(Total for question = 2 marks)

Q23.

This question is about the structure of atoms.

(a) State what is meant by the term orbital.

www.RocketRevise.com

www.RocketRevise.com

(2)

.....

.....

.....

.....

.....

(b) State the shape of an s orbital and the shape of a p orbital.

(1)

RocketRevise

.....

.....

(c) Describe what can be deduced about the electronic structure of sodium from its successive ionisation energies.

(3)

.....

.....

.....

.....

.....

.....

.....

(d) (i) The log of the **fourth** ionisation energy for six elements is shown in the table.

RocketRevise

www.RocketRevise.com

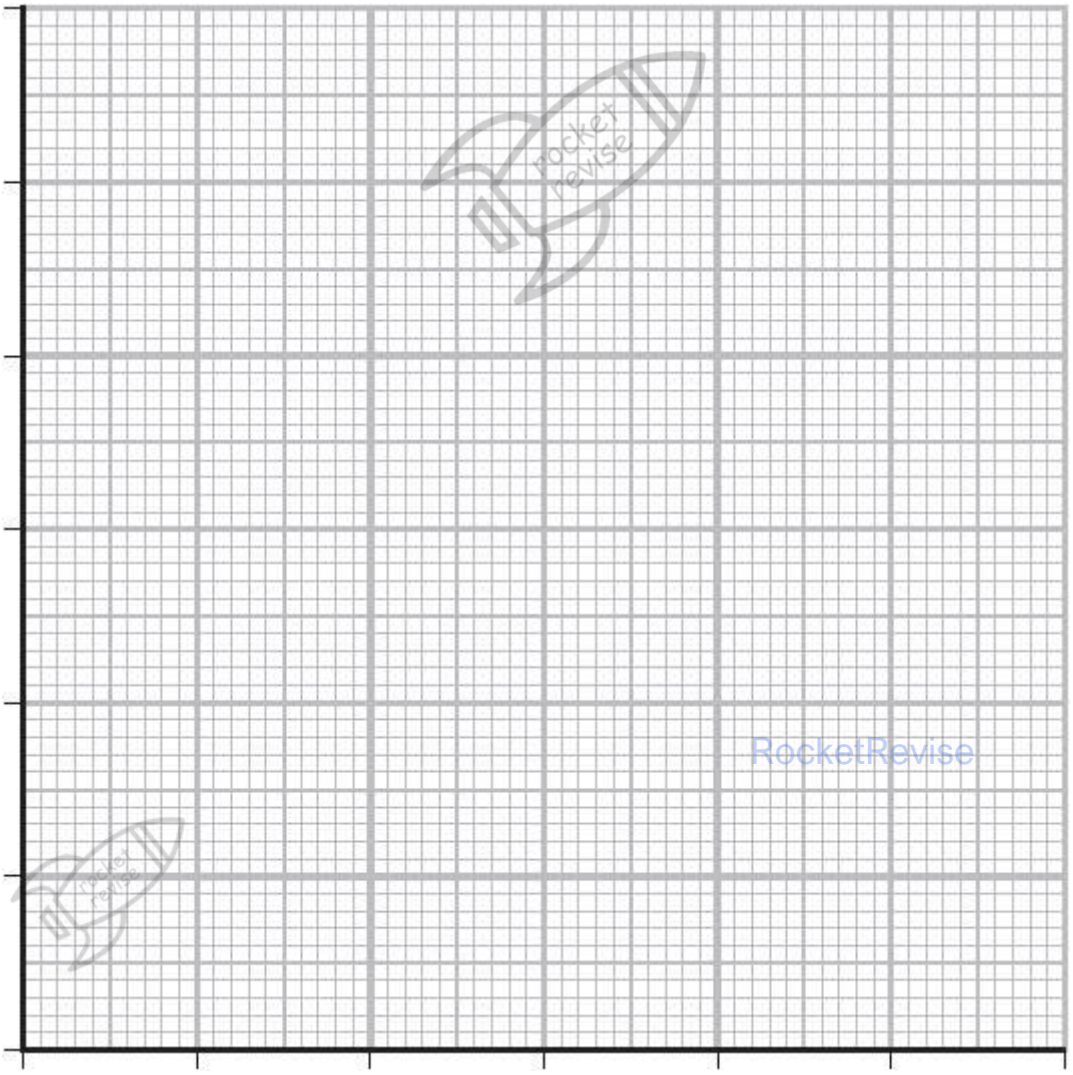
www.RocketRevise.com

Element	$\log_{10}$ (fourth ionisation energy / kJ mol^{-1})
O	3.87
F	3.92
Ne	3.97
Na	
Mg	4.02
Al	4.06

RocketRevise

Plot a graph of these data.

(3)



(ii) Use your graph to estimate the fourth ionisation energy for sodium, in kJ mol^{-1}.

(1)

www.RocketRevise.com

www.RocketRevise.com

(iii) Suggest why the fourth ionisation energies of neon and sodium are similar in magnitude even though the elements are in different periods of the Periodic Table.

(1)

.....

RocketRevise

.....

.....

.....

.....

(Total for question = 11 marks)

Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation represents the **third** ionisation energy for element J?

- ☒ **A** $J^{2+}(g) \rightarrow J^{3+}(g) + e^-$
- ☒ **B** $J^{3+}(g) \rightarrow J^{4+}(g) + e^-$
- ☒ **C** $J(g) \rightarrow J^{3+}(g) + e^-$
- ☒ **D** $J(g) \rightarrow J^{3+}(g) + 3e^-$

RocketRevise

(Total for question = 1 mark)

Q25.

This question is about iron and some of its compounds.

(a) Complete the table to show the numbers of subatomic particles in $^{56}Fe^{2+}$.

(1)

www.RocketRevise.com

www.RocketRevise.com

Number of protons	Number of neutrons	Number of electrons

(b) A sample of iron contains the following isotopes.

Isotope	Percentage abundance
^{54}Fe	5.84
^{56}Fe	91.68
^{57}Fe	2.17
^{58}Fe	0.31

RocketRevise

Calculate the relative atomic mass of this sample of iron.
Give your answer to **three** significant figures.

(2)

(c) Magnesium reacts with aqueous iron(II) sulfate in a displacement reaction.

Write the **ionic** equation for this reaction.
Include state symbols.

(2)

RocketRevise

(d) 25.00 g of a compound contains 6.98 g of iron and 6.03 g of sulfur.

The remaining mass is oxygen.
Calculate the **empirical** formula of this compound.
[A_r values: O = 16.0 S = 32.1 Fe = 55.8]

(3)

www.RocketRevise.com

www.RocketRevise.com

RocketRevise

(e) When 6.95 g of $FeSO_4.\mathbf{x}H_2O$ is heated, 2.00 g of iron(III) oxide, 0.80 g of sulfur dioxide and 1.00 g of sulfur trioxide

are produced.

The only other product is water.

Deduce the overall equation for the reaction using these data.
State symbols are not required.
You **must** show your working.
[A_r values: H = 1.0 O = 16.0 S = 32.1 Fe = 55.8]

(5)

RocketRevise

(Total for question = 13 marks)

www.RocketRevise.com

www.RocketRevise.com

Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which ion has the electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ in its ground state?

- ☒ **A** Al^{3+}
- ☒ **B** Cl^-
- ☒ **C** N^{3-}
- ☒ **D** Na^+

(Total for question = 1 mark)

RocketRevise

Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation represents the **second** ionisation energy of magnesium?

- ☒ **A** $Mg(g) \rightarrow Mg^{2+}(g) + 2e^-$
- ☒ **B** $Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$
- ☒ **C** $Mg(s) \rightarrow Mg^{2+}(s) + 2e^-$
- ☒ **D** $Mg^+(s) \rightarrow Mg^{2+}(s) + e^-$

Use this space for any rough working. Anything you write in this space will gain no credit.

RocketRevise

(Total for question = 1 mark)

Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The numbers of subatomic particles present in four species **W**, **X**, **Y** and **Z** are given in the table.

www.RocketRevise.com

www.RocketRevise.com

Species	Number of protons	Number of neutrons	Number of electrons
W	19	20	18
X	19	20	19
Y	20	20	18
Z	20	22	20

Which of these species are isotopic?

- ☒ **A** **W** and **X**
- ☒ **B** **W** and **Y**
- ☒ **C** **X** and **Z**
- ☒ **D** **Y** and **Z**

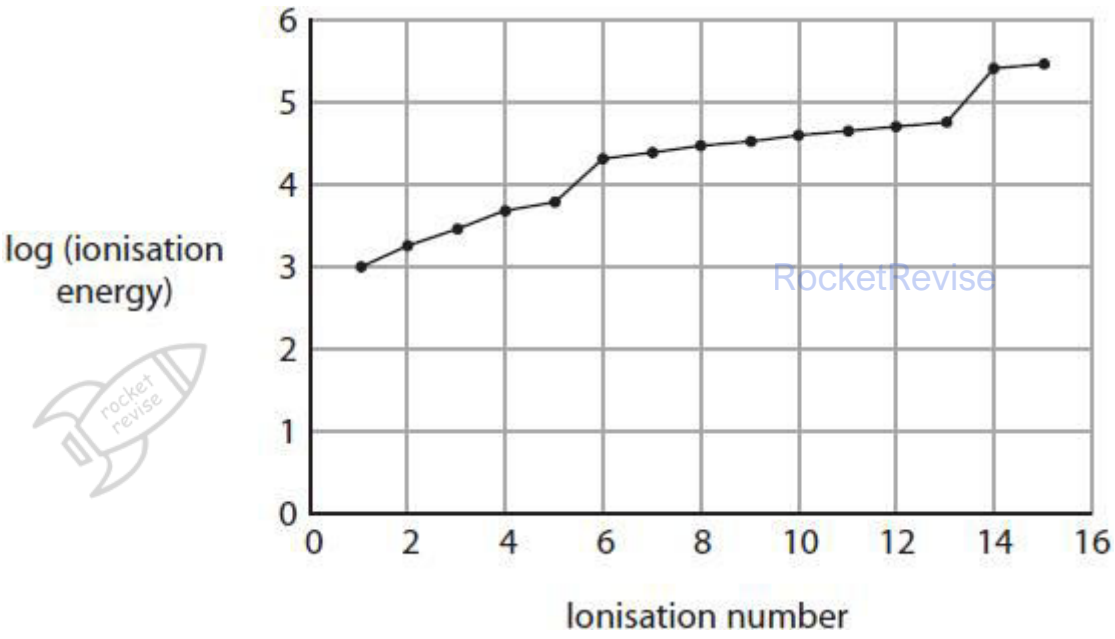
RocketRevise

(Total for question = 1 mark)

Q29.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The graph shows log (ionisation energy) against ionisation number for the successive ionisations of an element.



In this element, how many quantum shells contain electrons, and how many electrons are in the outer quantum shell?

www.RocketRevise.com

www.RocketRevise.com

	Number of quantum shells containing electrons	Number of electrons in the outer quantum shell
☒ A	3	2
☒ B	3	5
☒ C	5	2
☒ D	5	5

(Total for question = 1 mark)

RocketRevise

Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Iodine exists as one isotope with mass number 127.

Chlorine exists as two isotopes with mass numbers 35 and 37.

How many molecular ion (ICl_3^+) peaks are there in the mass spectrum of ICl_3?

☒ **A** 2
☒ **B** 3
☒ **C** 4
☒ **D** 5

(Total for question = 1 mark)

Q31.

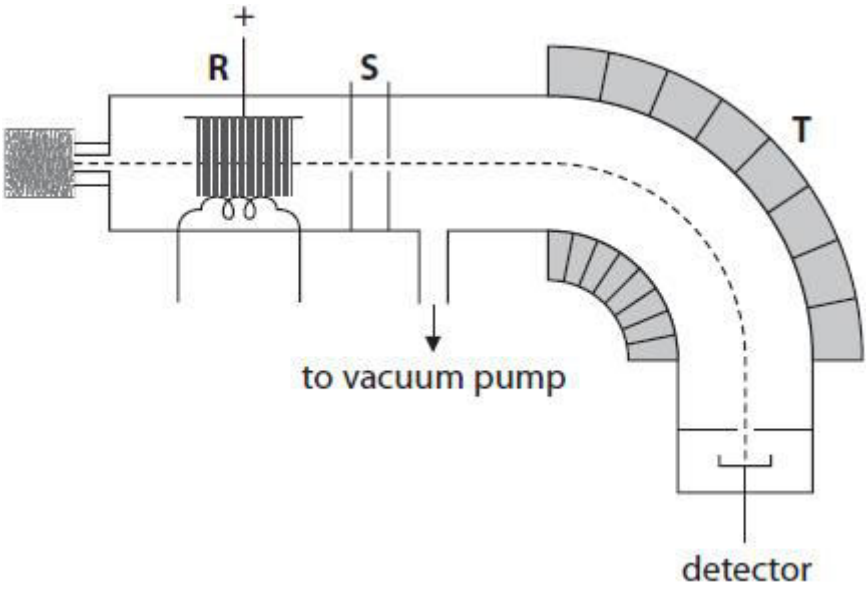
RocketRevise

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The diagram shows a mass spectrometer.

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

(a) Which process occurs in region **R**?

(1)

- ☒ **A** the sample is vaporised using a heater
- ☒ **B** electrons are removed from molecules or atoms and positive ions are formed
- ☒ **C** electrons are added to the molecules or atoms and negative ions are formed
- ☒ **D** ions are accelerated by an electric field

(b) Which statement is correct for region **T**?

(1)

- ☒ **A** ions with a greater mass have a smaller deflection
- ☒ **B** ions with a greater mass have a greater deflection
- ☒ **C** ions with a greater charge have a smaller deflection
- ☒ **D** ions are speeded up by a magnetic field

(Total for question = 2 marks)

Q32.

RocketRevise

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The **Period 2** element with the **highest** melting temperature is

- ☒ **A** aluminium
- ☒ **B** boron
- ☒ **C** carbon
- ☒ **D** silicon

www.RocketRevise.com

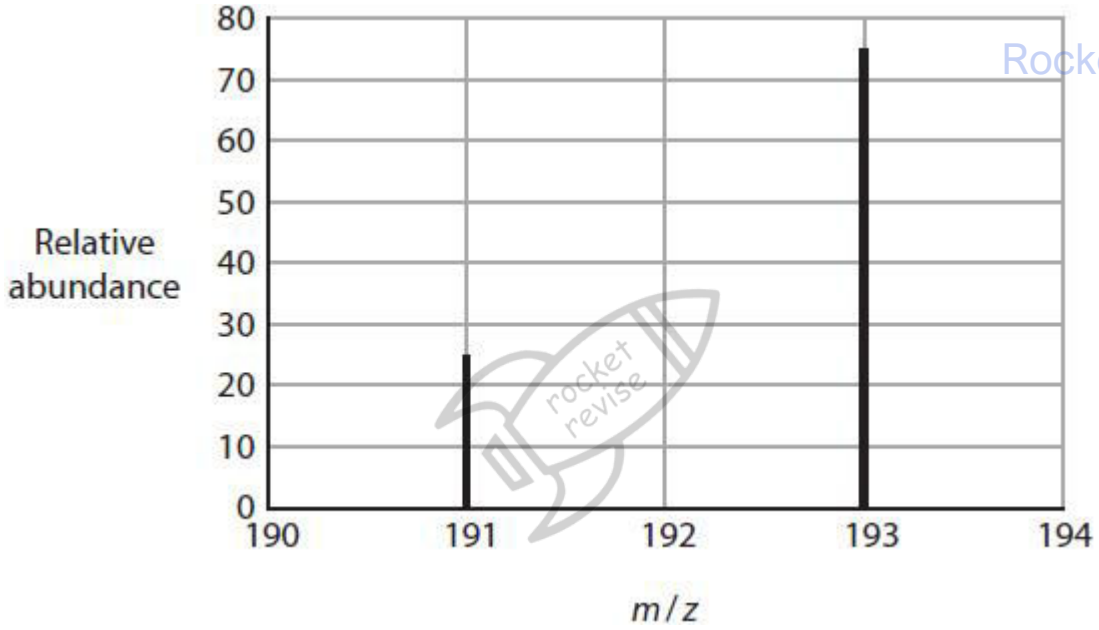
www.RocketRevise.com

(Total for question = 1 mark)

Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The mass spectrum of a sample of an element has only two peaks.



What is the approximate relative atomic mass of the element in this sample?

- ☐ **A** 191.5
- ☐ **B** 192.0
- ☐ **C** 192.5
- ☐ **D** 193.0

(Total for question = 1 mark)

Q34.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these atoms has the most neutrons?

www.RocketRevise.com

www.RocketRevise.com

☒ **A** $^{115}_{49}In$

☒ **B** $^{124}_{50}Sn$

☒ **C** $^{123}_{51}Sb$

☒ **D** $^{124}_{52}Te$

(Total for question = 1 mark)

Q35.

RocketRevise

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A stable ion, M^{3+}, contains 18 electrons.

In which block of the Periodic Table is element **M** found?

☒ **A** s

☒ **B** p

☒ **C** d

☒ **D** f

(Total for question = 1 mark)

Q36.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which equation represents the **second** ionisation energy of chlorine?

RocketRevise

☒ **A** $Cl^-(g) + e^- \rightarrow Cl^{2-}(g)$

☒ **B** $Cl(g) + 2e^- \rightarrow Cl^{2-}(g)$

☒ **C** $2Cl(g) \rightarrow 2Cl^+(g) + 2e^-$

☒ **D** $Cl^+(g) \rightarrow Cl^{2+}(g) + e^-$

(Total for question = 1 mark)

www.RocketRevise.com

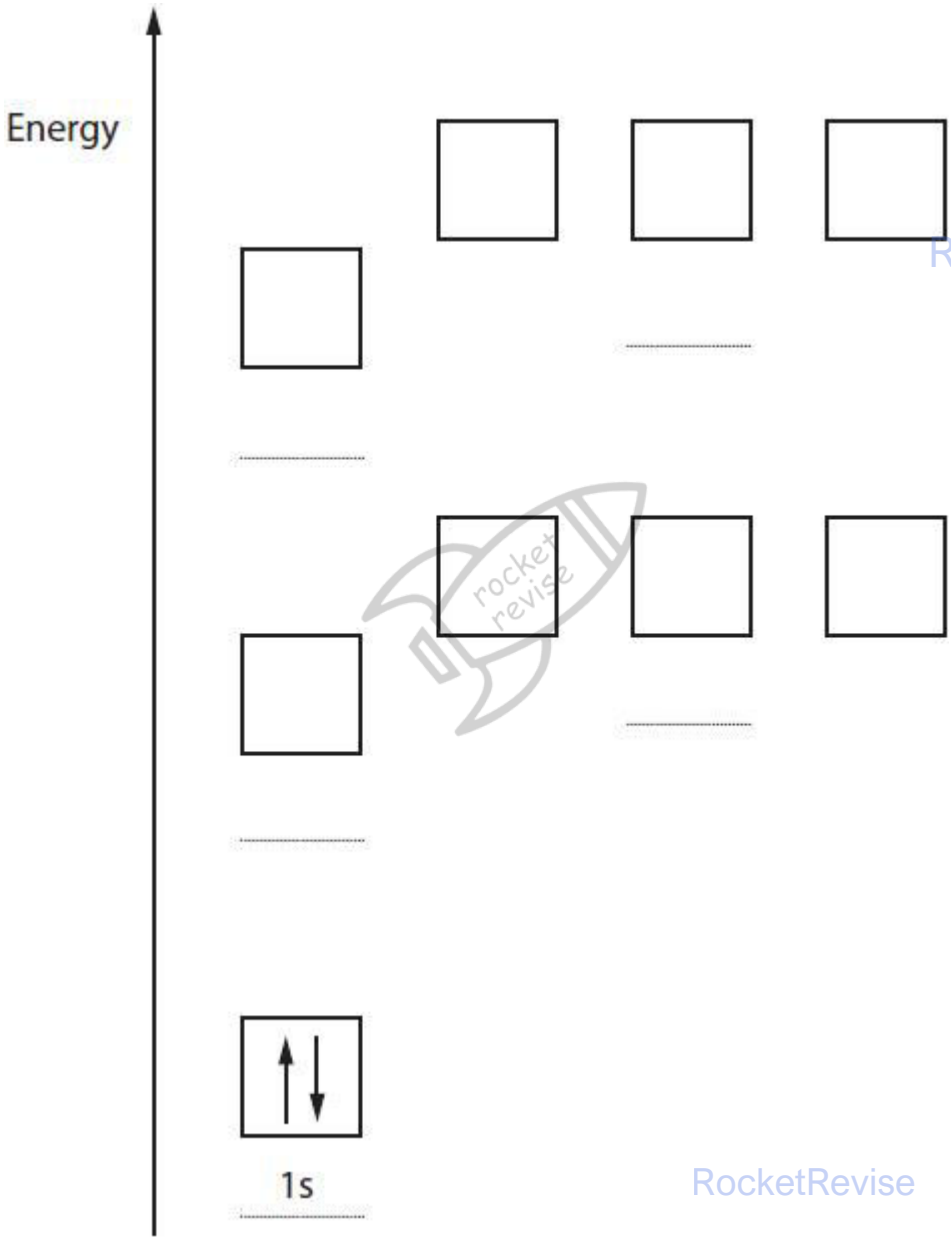
www.RocketRevise.com

Q37.

This question is about the element sulfur.

(a) Complete the diagram to show the electronic configuration for a sulfur atom in the ground state.
Include labels for each subshell.

(2)



RocketRevise

RocketRevise

(b) Write an equation for the **first** ionisation energy of sulfur.
Include state symbols.

(2)

(c) Explain why the first ionisation energy of sulfur is less than the first ionisation energies of **both** phosphorus and chlorine.

(3)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(d) A sample of sulfur contains four isotopes.

Isotope	^{32}S	^{33}S	^{34}S	^{36}S
Percentage abundance	94.88	0.83	4.27	0.02

(i) State what is meant by the term **isotopes**, in terms of subatomic particles.

(2)

.....

.....

.....

(ii) Calculate the relative atomic mass of sulfur in this sample.
Give your answer to **two** decimal places.

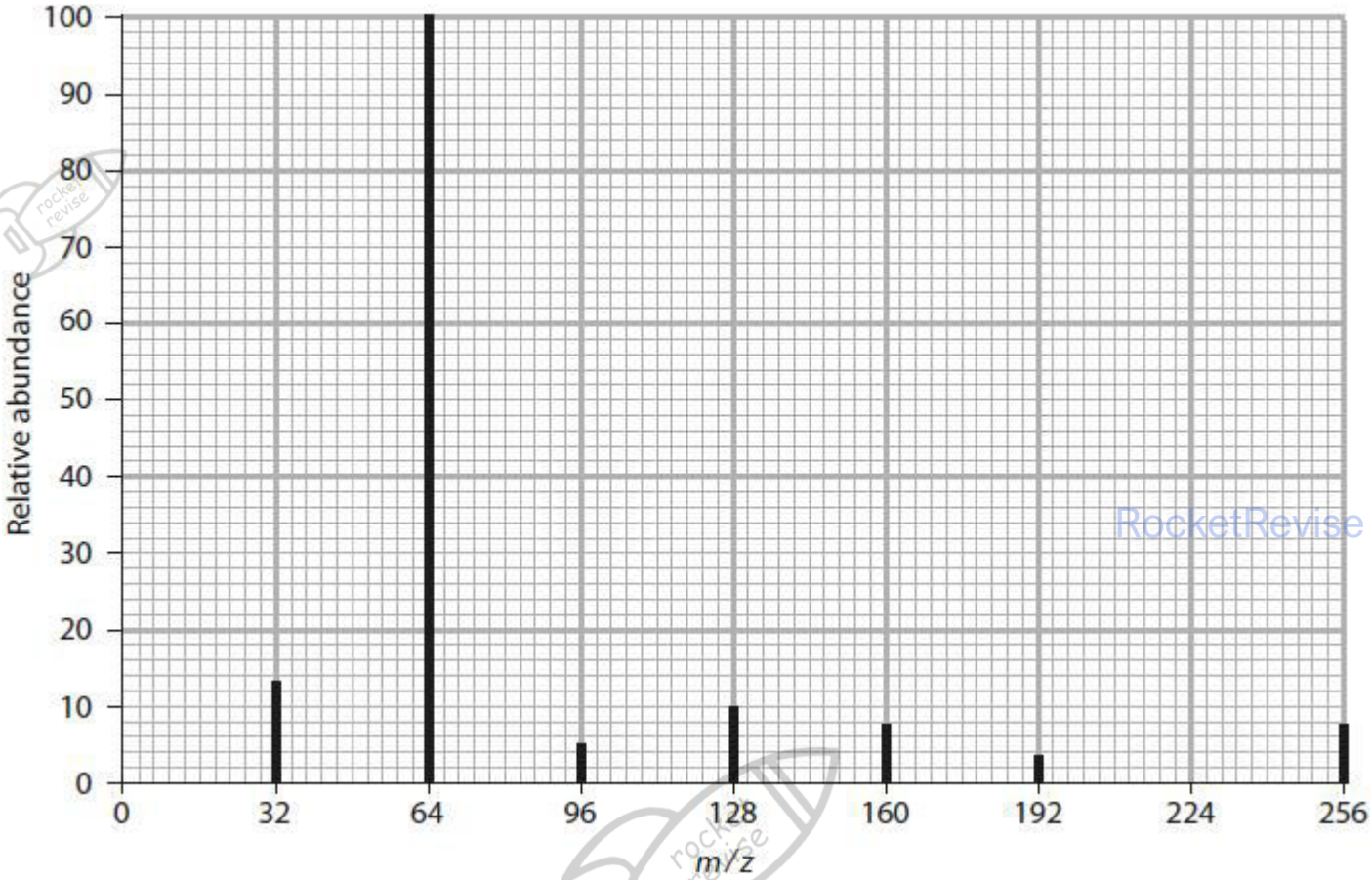
(2)

(e) The mass spectrum of a sample of sulfur with ^{32}S as the only isotope is shown.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



(i) Calculate the number of sulfur atoms in the molecular ion.
You **must** show your working.

(1)

(ii) Suggest the **formula** of the **most stable ion** shown by this spectrum.

(2)

.....

(Total for question = 14 marks)

Q38.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which series are the elements in order of **increasing** melting temperature?

www.RocketRevise.com

www.RocketRevise.com

☒ **A** $I_2 < Br_2 < Cl_2 < F_2$

☒ **B** Li < Be < B < C

☒ **C** Li < Na < K < Rb

☒ **D** Si < P < S < Cl

(Total for question = 1 mark)

Q39.

RocketRevise

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the total number of orbitals in a sulfur atom which contain at least one electron?

☒ **A** 3

☒ **B** 5

☒ **C** 8

☒ **D** 9

(Total for question = 1 mark)

Q40.

This question is about copper and its compounds.

(a) Complete the electronic configurations of Cu and Cu^{2+}.

(2)

Cu [Ar] .....

Cu^{2+} [Ar] .....

RocketRevise

(b) A sample of copper contains the isotopes ^{63}Cu and ^{65}Cu.

(i) Complete the table to show the numbers of subatomic particles in the atoms of these two isotopes of copper.

(2)

Isotope	Protons	Neutrons	Electrons
^{63}Cu			
^{65}Cu			

www.RocketRevise.com

www.RocketRevise.com

(ii) Explain the term isotopes, using the information in the table.

(2)

.....

.....

.....

.....

.....

(iii) State why the two isotopes of copper have the same chemical reactions.

(1)

.....

RocketRevise

.....

(iv) The relative atomic mass of copper in this sample is 63.4.
Calculate the percentage abundances of the isotopes ^{63}Cu and ^{65}Cu in this sample.
You must show your working.

(2)

(c) Copper(II) sulfate, CuSO4, can be made by reacting solid copper(II) carbonate with dilute sulfuric acid.

(i) Write an equation for the reaction that occurs.
State symbols are not required.

(1)

RocketRevise

(ii) An experiment was carried out to produce pure, dry crystals of hydrated copper(II) sulfate, $CuSO_4 \cdot 5H_2O$.
Copper(II) carbonate was mixed with 50.0 cm^3 of 1.00 mol dm^{-3} sulfuric acid until no more reacted.
The mass of $CuSO_4 \cdot 5H_2O$ obtained was 10.87 g.
Calculate the percentage yield for this reaction, giving your answer to an appropriate number of significant figures.

www.RocketRevise.com

www.RocketRevise.com

[Molar mass of $CuSO_4 \cdot 5H_2O$ = 249.6 g mol^{-1}]

(4)

RocketRevise

(Total for question = 14 marks)

Q41.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

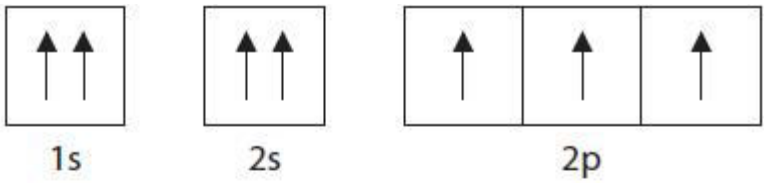
What is the electronic configuration of a nitrogen atom?

RocketRevise

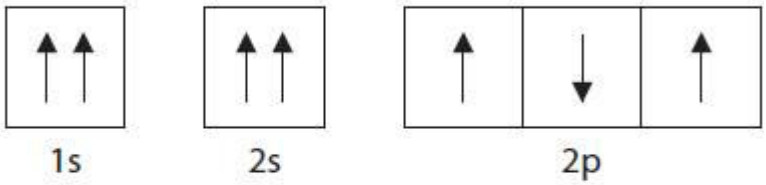
www.RocketRevise.com

www.RocketRevise.com

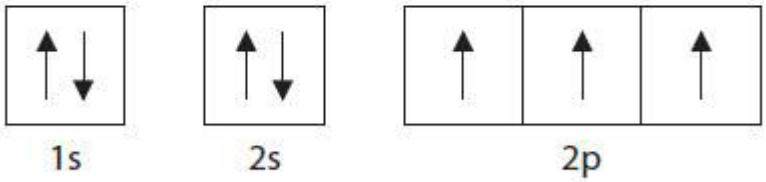
☒ **A**



☒ **B**

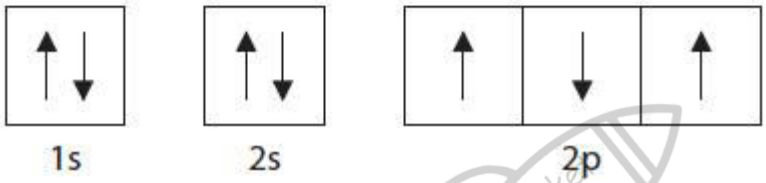


☒ **C**



RocketRevise

☒ **D**



(Total for question = 1 mark)

Q42.

Which equation represents the first ionisation energy of iodine?

☒ **A** $I_2(s) \rightarrow 2I^+(g) + 2e^-$

☒ **B** $I_2(g) \rightarrow 2I^+(g) + 2e^-$

☒ **C** $\frac{1}{2}I_2(s) \rightarrow I^+(g) + e^-$

☒ **D** $I(g) \rightarrow I^+(g) + e^-$

Use this space for any rough working. Anything you write in this space will gain no credit.

RocketRevise

(Total for question = 1 mark)

Q43.

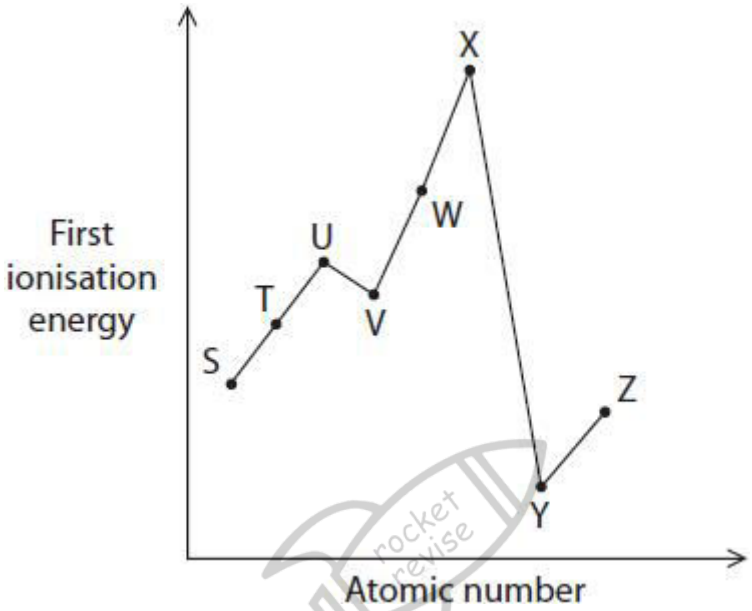
www.RocketRevise.com

www.RocketRevise.com

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The graph shows the first ionisation energies of some consecutive elements from Periods 2 and 3 of the Periodic Table.

The letters used to label the elements are **not** their chemical symbols.



RocketRevise

(a) Which element has only **one** p orbital containing a pair of electrons?

(1)

- ☐ **A** element U
- ☐ **B** element V
- ☐ **C** element W
- ☐ **D** element X

(b) Which element is the least reactive?

(1)

- ☐ **A** element S
- ☐ **B** element X
- ☐ **C** element Y
- ☐ **D** element Z

RocketRevise

(c) Which element has the greatest **second** ionisation energy?

(1)

- ☐ **A** element S
- ☐ **B** element T
- ☐ **C** element X
- ☐ **D** element Y

(Total for question = 3 marks)

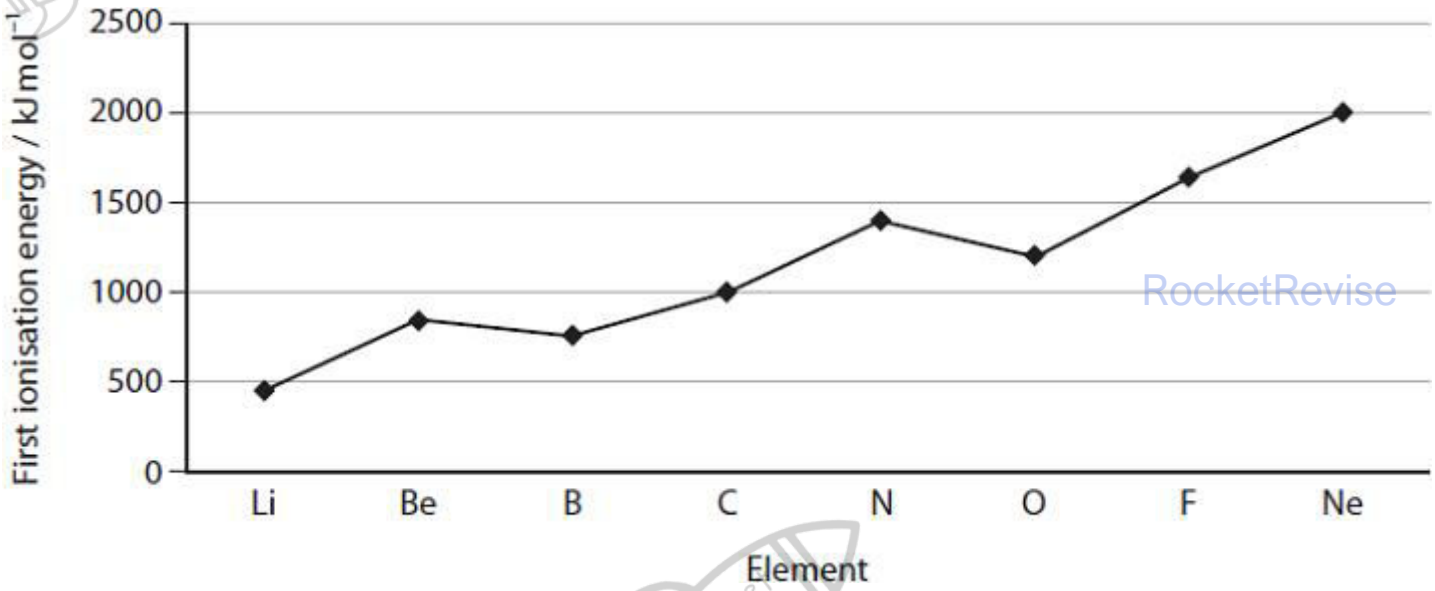
www.RocketRevise.com

www.RocketRevise.com

Q44.

This question is about the ionisation energies of the elements in Period 2 of the Periodic Table.

(a) The first ionisation energies of the Period 2 elements are shown.



(i) Give an equation that represents the first ionisation energy of lithium. Include state symbols.

(1)

(ii) Explain why there is a general increase in the first ionisation energy across the period.

(2)

.....

.....

.....

.....

.....

.....

(iii) Explain why the first ionisation energy of oxygen is lower than that of nitrogen.

(2)

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

(b) All the successive ionisation energies of nitrogen are shown in the table.

Ionisation number	1	2	3	4	5	6	7
Ionisation energy / kJ mol^{-1}	1402	2856	4578	7475	9445	53 267	64 360

Explain the trend in the successive ionisation energies of nitrogen.

(2)

.....

.....

.....

.....

.....

.....

(Total for question = 7 marks)

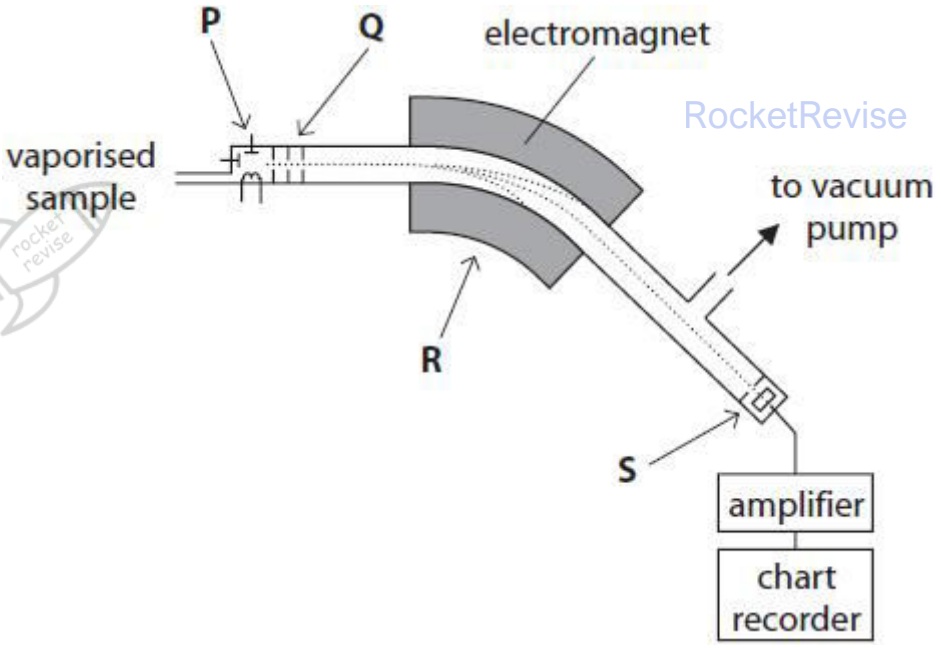
Q45.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Mass spectrometers are used in chemical analysis.

(a) In which region of this mass spectrometer are the ions accelerated?

(1)

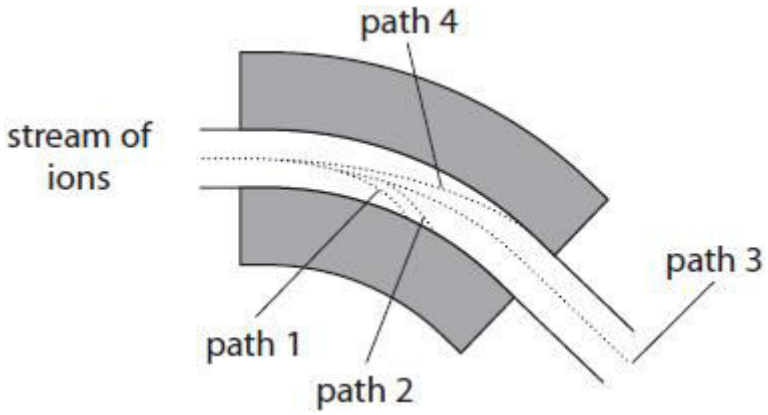


www.RocketRevise.com

www.RocketRevise.com

- **A** region **P**
- **B** region **Q**
- **C** region **R**
- **D** region **S**

(b) Four iron ions are present in a stream of ions in a mass spectrometer. The paths of these ions are shown.



RocketRevise

Which of these ions will follow path 1 through the mass spectrometer?

(1)

- **A** $^{54}Fe^{+}$
- **B** $^{54}Fe^{2+}$
- **C** $^{56}Fe^{+}$
- **D** $^{56}Fe^{2+}$

(c) Naturally-occurring chlorine has two isotopes, ^{35}Cl and ^{37}Cl.

How many peaks due to the singly-charged **molecular** ions are seen in the mass spectrum of chlorine, Cl_2?

(1)

- **A** 1
- **B** 2
- **C** 3
- **D** 4

(Total for question = 3 marks)

Use this space for any rough working. Anything you write in this space will gain no credit.

Q46.

www.RocketRevise.com

www.RocketRevise.com

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The element manganese has the atomic number $Z = 25$.

What are the numbers of s, p and d electrons in an atom of manganese?

	s electrons	p electrons	d electrons
☐ **A**	6	12	7
☐ **B**	8	12	5
☐ **C**	6	18	1
☐ **D**	8	17	0

RocketRevise

Use this space for any rough working. Anything you write in this space will gain no credit.

(Total for question = 1 mark)

Q47.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The atomic number of the element scandium is 21 and the mass number of its only isotope is 45.

RocketRevise

(a) What is the number of electrons in a scandium ion, Sc^+ ?

(1)

☐ **A** 20
☐ **B** 21
☐ **C** 22
☐ **D** 23

(b) In a mass spectrometer, scandium forms Sc^+ and Sc^{2+} ions.

What is the m / z value for the mass spectrum peak due to the Sc^{2+} ions?

(1)

☐ **A** 22.5
☐ **B** 33.0

www.RocketRevise.com

www.RocketRevise.com

☒ **C** 45.0
☒ **D** 90.0

(Total for question = 2 marks)

Q48.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

RocketRevise

An atom of $^{15}_{7}N$ forms a nitride ion, N^{3-}.

What are the numbers of protons, neutrons and electrons in this ion?

	Protons	Neutrons	Electrons
☒ **A**	7	8	7
☒ **B**	7	8	4
☒ **C**	7	8	10
☒ **D**	8	7	10

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

RocketRevise

Q49.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the electronic configuration of an oxygen atom in its ground state?

www.RocketRevise.com

www.RocketRevise.com

	1s	2s	$2p_x$	$2p_y$	$2p_z$	3s
☒ A	↑↓	↑↓	↑↓	↑↓		
☒ B	↑↓	↑↓	↑↑	↑	↑	
☒ C	↑↓	↑↓	↑	↑	↑	↑
☒ D	↑↓	↑↓	↑↓	↑	↑	

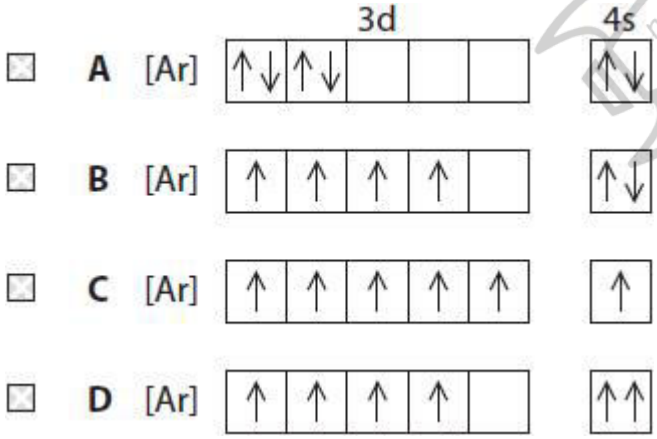
(Total for question = 1 mark)

RocketRevise

Q50.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the electronic configuration of chromium?



(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

RocketRevise

Q51.

www.RocketRevise.com

www.RocketRevise.com

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is a reason why fluorine has a higher first ionisation energy than oxygen?

- ☒ **A** a fluorine atom has fewer unpaired electrons
- ☒ **B** a fluorine atom has fewer shells of electrons
- ☒ **C** a fluorine atom has more electrons
- ☒ **D** a fluorine atom has more protons

(Total for question = 1 mark)

RocketRevise

Q52.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the equation for the **second** ionisation of element J?

- ☒ **A** J(g) → $J^{2+}(g) + 2e^-$
- ☒ **B** $J^+(g)$ → $J^{2+}(g) + e^-$
- ☒ **C** $J^+(g)$ → $J^{3+}(g) + 2e^-$
- ☒ **D** $J^{2+}(g)$ → $J^{3+}(g) + e^-$

(Total for question = 1 mark)

Q53.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many neutrons and electrons are there in $^{190}_{79}Au^{2+}$?

RocketRevise

	Neutrons	Electrons
☒ **A**	111	77
☒ **B**	111	81
☒ **C**	190	79
☒ **D**	190	77

(Total for question = 1 mark)

www.RocketRevise.com

www.RocketRevise.com

Q54.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The first seven ionisation energies, in kJ mol^{-1}, of an element are shown.

1010, 1900, 2910, 4960, 6270, 21 300, 25 400

In which group of the Periodic Table is this element located?

- ☐ **A** Group 3
- ☐ **B** Group 4
- ☐ **C** Group 5
- ☐ **D** Group 6

RocketRevise

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

Q55.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The first ionisation energy of sulfur is lower than that of phosphorus.

RocketRevise

Which is the best explanation for this?

- ☐ **A** the atomic radius of sulfur is greater than that of phosphorus
- ☐ **B** the electronegativity of sulfur is greater than that of phosphorus
- ☐ **C** the repulsion between the outer electrons of sulfur is greater than that of phosphorus
- ☐ **D** the shielding by the inner shell electrons of sulfur is greater than that of phosphorus

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

www.RocketRevise.com

www.RocketRevise.com

Q56.

This question is about minerals containing metallic elements.

(a) The green mineral zaratite contains basic nickel(II) carbonate, $Ni_3CO_3(OH)_4 \cdot 4H_2O$.

Basic nickel(II) carbonate can be formed by the reaction of nickel(II) sulfate with sodium carbonate under alkaline conditions, as shown.

$$3NiSO_4(aq) + Na_2CO_3(aq) + 4NaOH(aq) + 4H_2O(l) \rightarrow Ni_3CO_3(OH)_4 \cdot 4H_2O(s) + 3Na_2SO_4(aq)$$

RocketRevise

[A_r values: H = 1.0 C = 12.0 O = 16.0 Na = 23.0 S = 32.1 Ni = 58.7]

(i) Calculate the relative formula mass of basic nickel(II) carbonate, $Ni_3CO_3(OH)_4 \cdot 4H_2O$.

(1)

(ii) Calculate the atom economy by mass for the formation of basic nickel(II) carbonate by the reaction shown.

(2)

(iii) Give the reason why use of the term 'relative molecular mass' is not appropriate in (a)(i).

(1)

.....

.....

.....

.....

RocketRevise

(b) Chemists often simplify full chemical equations into ionic equations by removing spectator ions.

(i) Give the electronic configuration of the nickel ion present in nickel(II) compounds.

(1)

.....

(ii) Write the **ionic** equation for the formation of basic nickel(II) carbonate from nickel(II) sulfate. State symbols are not required.

$$3NiSO_4(aq) + Na_2CO_3(aq) + 4NaOH(aq) + 4H_2O(l) \rightarrow Ni_3CO_3(OH)_4 \cdot 4H_2O(s) + 3Na_2SO_4(aq)$$

(1)

www.RocketRevise.com

www.RocketRevise.com

(c) Another basic carbonate found in minerals contains the metal element **X** and is formed by the reaction shown.

$$2\mathbf{X}SO_4 + 2Na2CO_3 + H_2O \rightarrow \mathbf{X}_2CO_3(OH)_2 + 2Na_2SO_4 + CO_2$$

1.995 g of $\mathbf{X}SO_4$ is reacted with excess Na_2CO_3 forming 150.0 cm^3 of carbon dioxide gas, CO_2, at room temperature and pressure, r.t.p.
[Molar volume of a gas at r.t.p. = 24.0 $dm^3\ mol^{-1}$]

(i) Calculate the relative formula mass of $\mathbf{X}SO_4$, using these data.

(4)

(ii) Deduce the identity of **X**, using your answer to (c)(i).

(1)

(Total for question = 11 marks)

Q57.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

An atom of $^{27}_{13}Al$ forms the ion, Al^{3+}.
What are the numbers of protons, electrons and neutrons in this ion?

www.RocketRevise.com

www.RocketRevise.com

	protons	electrons	neutrons
☒ **A**	13	13	14
☒ **B**	10	13	14
☒ **C**	14	10	13
☒ **D**	13	10	14

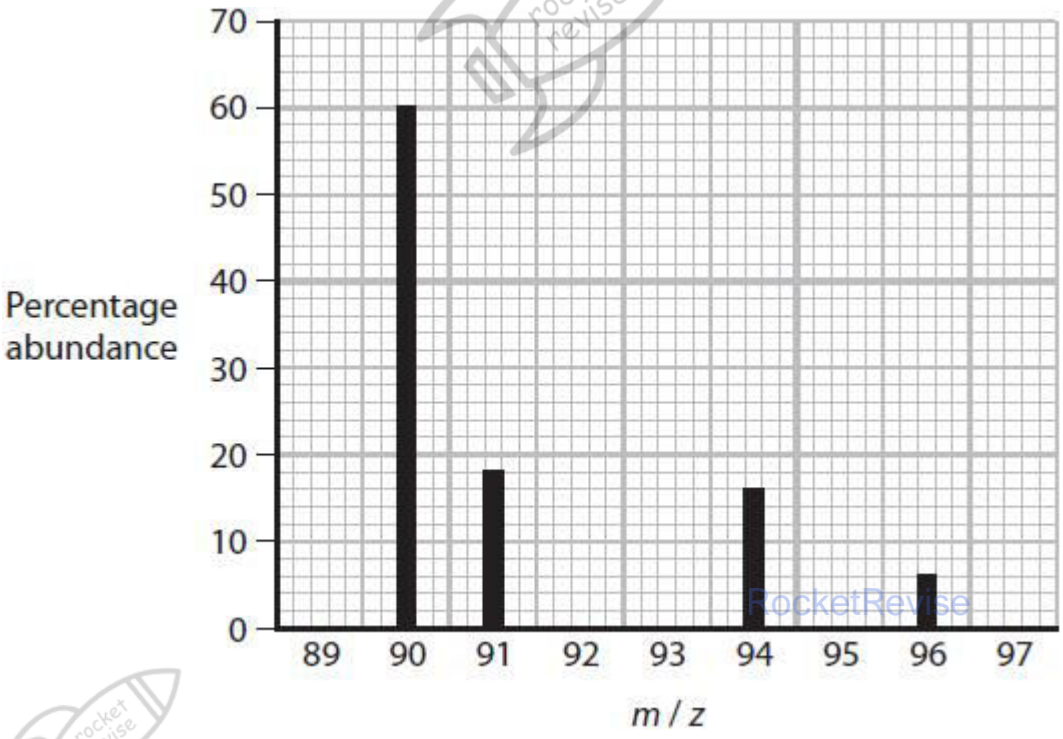
(Total for question = 1 mark)

RocketRevise

Q58.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The mass spectrum of an element is shown.



What is the relative atomic mass of this element?

☒ **A** 90.0
☒ **B** 90.9
☒ **C** 91.2
☒ **D** 92.8

(Total for question = 1 mark)

www.RocketRevise.com

www.RocketRevise.com

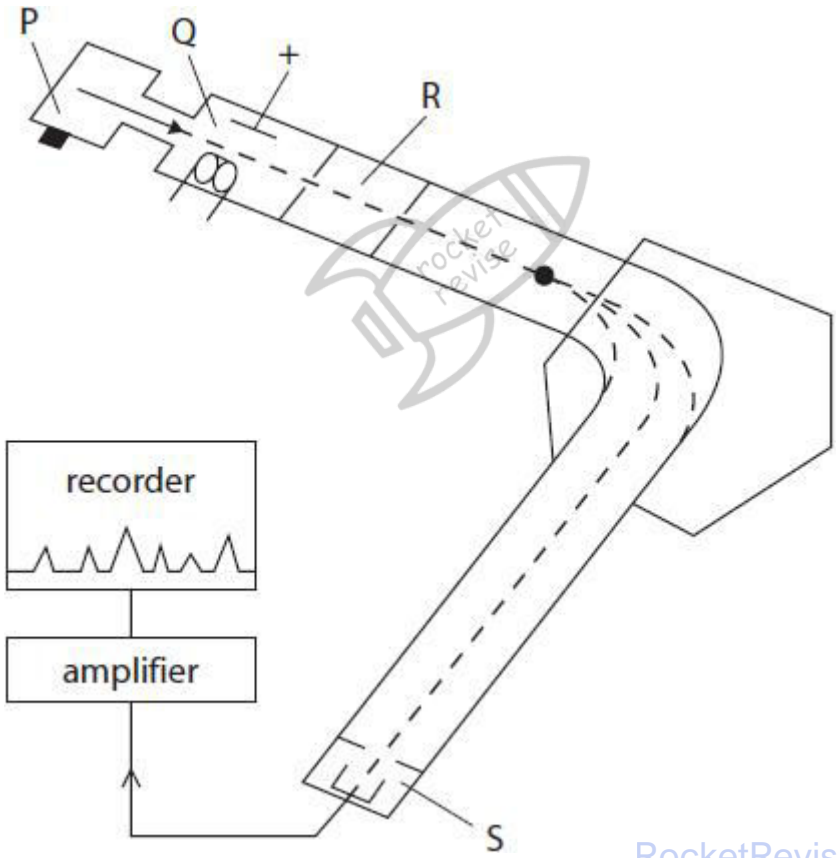
Use this space for any rough working. Anything you write in this space will gain no credit.

Q59.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

RocketRevise

In which region of a mass spectrometer are particles ionised?



RocketRevise

- ☒ **A** region P
- ☒ **B** region Q
- ☒ **C** region R
- ☒ **D** region S

(Total for question = 1 mark)

Q60.

www.RocketRevise.com

www.RocketRevise.com

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A sample of the element chlorine, Cl_2, was analysed in a mass spectrometer. Chlorine has **two** isotopes.

What is the **total** number of peaks, due to ions with a single positive charge, which could be seen in the mass spectrum?

- ☒ **A** two
- ☒ **B** four
- ☒ **C** five
- ☒ **D** six

(Total for question = 1 mark)

Q61.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

What is the shape of a p orbital and the maximum number of electrons it can hold?

	shape	maximum number of electrons
☒ **A**		2
☒ **B**		6
☒ **C**		2
☒ **D**		6

(Total for question = 1 mark)

Q62.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

www.RocketRevise.com

www.RocketRevise.com

The electronic configurations of the atoms of four elements are shown.

What is the electronic configuration of the atom of element which has the **lowest** first ionisation energy?

- ☒ **A** $1s^2 2s^2 2p^6 3s^2$
- ☒ **B** $1s^2 2s^2 2p^6 3s^2 3p^1$
- ☒ **C** $1s^2 2s^2 2p^6 3s^2 3p^2$
- ☒ **D** $1s^2 2s^2 2p^6 3s^2 3p^3$

(Total for question = 1 mark)

RocketRevise

Q63.

This question is about bromine.

(a) Complete the electronic configuration for a bromine atom, using the s, p, d notation.

(1)

[Ar] .....

(b) Bromine exists as two isotopes with mass numbers 79 and 81.

(i) Complete the table to show the numbers of subatomic particles in a ^{79}Br atom and a $^{81}Br^-$ ion.

(2)

Species	Protons	Neutrons	Electrons
^{79}Br			
$^{81}Br^-$			

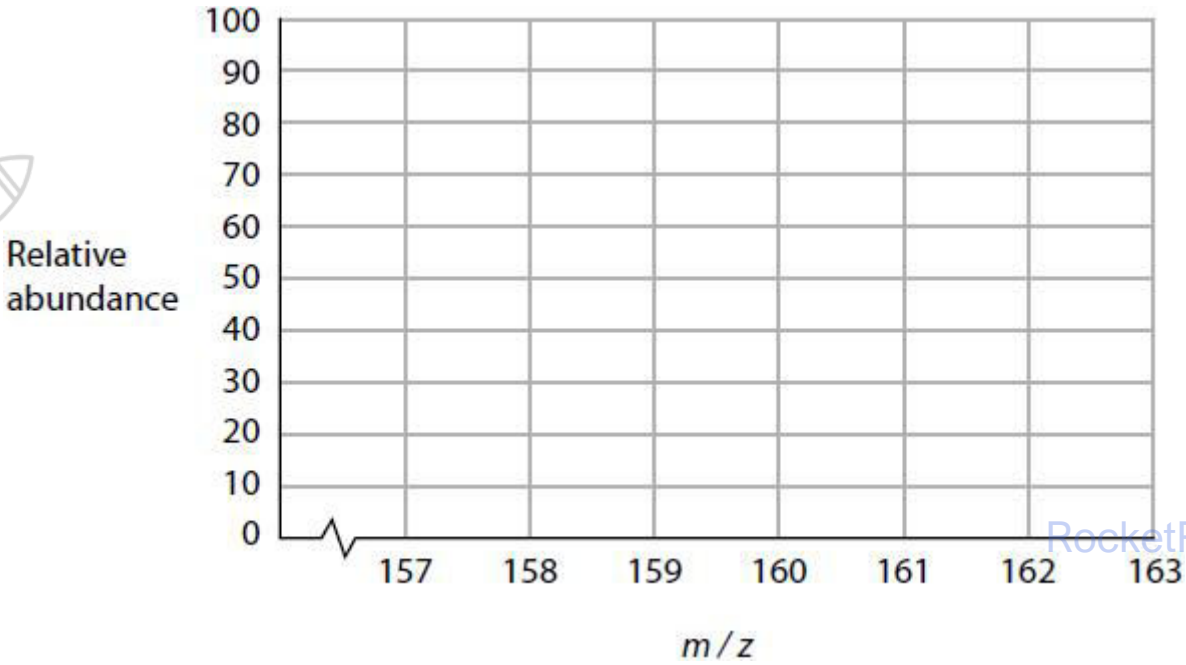
(ii) A sample of bromine contained equal amounts of the two isotopes.
Complete the mass spectrum to show the peaks you would expect for Br_2^+ from this sample of bromine gas.

(2)

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

(iii) Calculate the number of bromine molecules in 2.00 g of Br_2.
[Avogadro constant = 6.02×10^{23} mol^{-1}]

(2)

rocket revise

Number of molecules = ...

(c) A sample of bromine gas occupied 200 cm^3 at a temperature of 77 °C and a pressure of 1.51×10^5 Pa.

Calculate, using the ideal gas equation, the amount in moles of bromine molecules in this sample.
[$pV = nRT$ $R = 8.31$ J mol^{-1} K^{-1}]

(4)

RocketRevise

rocket revise

Amount of bromine molecules = ... mol

rocket revise

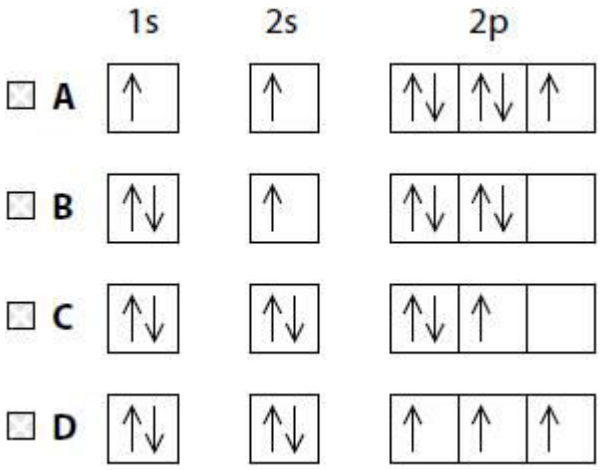
www.RocketRevise.com

www.RocketRevise.com

(Total for question = 11 marks)

Q64.

Which is the electronic configuration of nitrogen?



RocketRevise

(Total for question = 1 mark)

Q65.

What are the maximum numbers of electrons in a 2p orbital and in the third quantum shell?

	Maximum number of electrons in a 2p orbital	Maximum number of electrons in the third quantum shell
A	2	8
B	2	18
C	6	8
D	6	18

RocketRevise

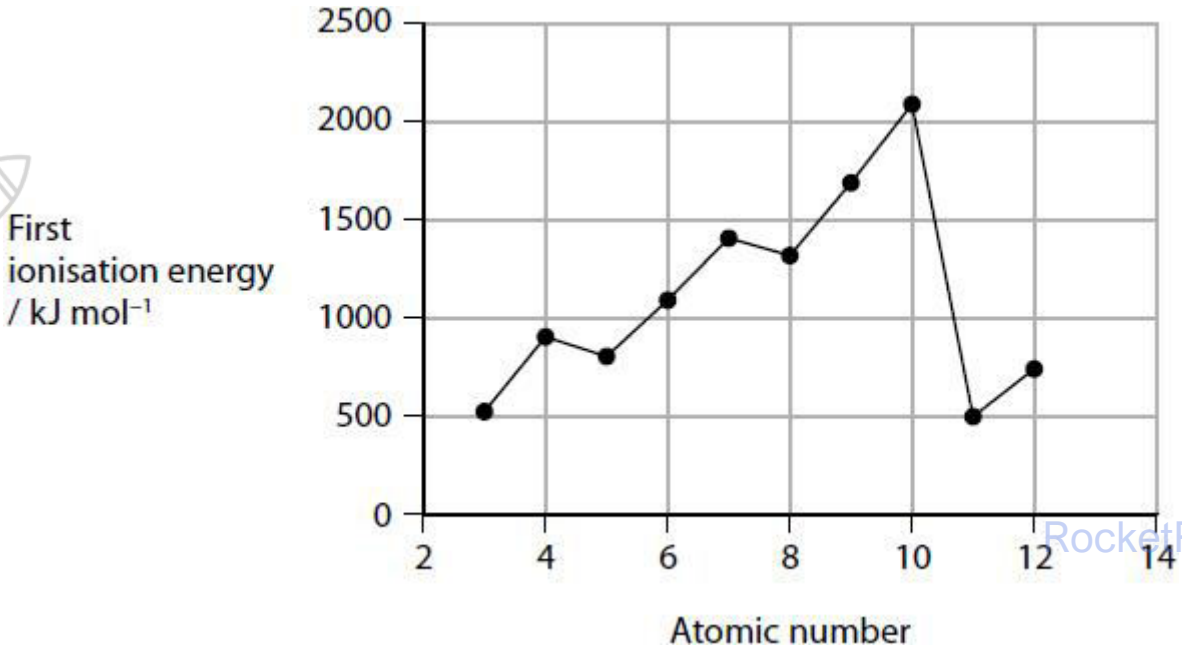
(Total for question = 1 mark)

Q66.

The graph shows the first ionisation energies for the elements with atomic numbers from 3 to 12.

www.RocketRevise.com

www.RocketRevise.com



(a) Write the equation for the first ionisation energy of nitrogen. Include state symbols.

(2)

(b) Explain the changes in first ionisation energy for the elements with atomic numbers from 3 to 10.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

(c) Explain why the first ionisation energy of element 11 is lower than that of element 3.

(2)

.....

.....

.....

.....

(Total for question = 8 marks)

www.RocketRevise.com

www.RocketRevise.com

Q67.

Data from the mass spectrum of a sample of pure iron is given in the table.

m / z	Relative peak height
28	0.1
54	6.3
56	100.0
57	2.4
58	0.3

RocketRevise

Which species is most likely to cause the peak at $m/z = 28$?

(1)

- ☒ **A** $^{28}Fe^{+}$
- ☒ **B** $^{56}Fe^{2+}$
- ☒ **C** $^{28}Si^{+}$
- ☒ **D** $^{84}Sr^{3+}$

(Total for question = 1 mark)

Q68.

This question is about phosphorus and sulfur.

Which species contains 15 protons, 16 neutrons and 18 electrons?

(1)

- ☒ **A** P^{3-}
- ☒ **B** P^{3+}
- ☒ **C** S^{2-}
- ☒ **D** S^{2+}

RocketRevise

(Total for question = 1 mark)

Q69.

A sample of neon contains the following isotopes.

www.RocketRevise.com

www.RocketRevise.com

Isotope	Percentage abundance
^{20}Ne	90.92
^{21}Ne	0.26
^{22}Ne	8.82

What is the relative atomic mass of neon to two decimal places?

(1)

- ☒ **A** 20.00
- ☒ **B** 20.09
- ☒ **C** 20.18
- ☒ **D** 21.00

RocketRevise

(Total for question = 1 mark)

Q70.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The table shows some data about fundamental particles in an atom.

Particle	Mass / g
electron	0.0009×10^{-24}
neutron	1.6748×10^{-24}
proton	1.6725×10^{-24}

RocketRevise

(a) An atom of hydrogen can be represented as ^{1}H.

What is the mass, in grams, of this hydrogen atom?

(1)

- ☒ **A** 1.6725×10^{-24}
- ☒ **B** 1.6734×10^{-24}
- ☒ **C** 3.3473×10^{-24}
- ☒ **D** 3.3482×10^{-24}

(b) Which of the particles would be deflected by an electric field?

(1)

- ☒ **A** electrons only
- ☒ **B** electrons and protons only

www.RocketRevise.com

www.RocketRevise.com

☒ **C** neutrons and protons only
☒ **D** electrons, neutrons and protons

(Total for question = 2 marks)

Q71.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

RocketRevise

This question is about electronic configurations.

(a) What is the electronic configuration of an atom of calcium?

(1)

☒ **A** [Ne] $3s^1$
☒ **B** [Ar] $4s^1$
☒ **C** [Ne] $3s^2$
☒ **D** [Ar] $4s^2$

(b) What is the electronic configuration of a Cr^{3+} ion?

(1)

☒ **A** [Ar] $3d^2 4s^1$
☒ **B** [Ar] $3d^3$
☒ **C** [Ar] $3d^4$
☒ **D** [Ar] $3d^5$

(Total for question = 2 marks)

Q72.

RocketRevise

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Each response gives the atomic numbers of two elements.

Which pair of atomic numbers are those of elements that are in different blocks of the Periodic Table?

☒ **A** 5, 9
☒ **B** 10, 16
☒ **C** 13, 18
☒ **D** 16, 20

(Total for question = 1 mark)

www.RocketRevise.com

www.RocketRevise.com

Q73.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which species does **not** contain a total of 16 neutrons?

- ☒ **A** a molecule of ethene, $^{12}C_2{}^{1}H_4$
- ☒ **B** a molecule of oxygen, $^{16}O_2$
- ☒ **C** an atom of silicon, ^{30}Si
- ☒ **D** an ion of sulfur, $^{32}S^{2-}$

RocketRevise

(Total for question = 1 mark)

Q74.

This question is about silicon.

(a) (i) Define relative atomic mass.

(2)

.....

.....

.....

.....

(ii) Calculate the relative atomic mass of a sample of silicon, using the isotopic abundance data provided.

Give your answer to 3 significant figures.

Isotope	Abundance (%)
^{28}Si	91.07
^{29}Si	4.62
^{30}Si	3.00
^{32}Si	1.31

RocketRevise

www.RocketRevise.com

www.RocketRevise.com

(2)

(iii) In the mass spectrum of silicon, there is also a small peak at $m / z = 14$.
Deduce the formula of this particle.

(1)

RocketRevise

(b) Consider the elements Al, Si, P and S.

(i) Explain the trend in the first ionisation energies of Al, Si and P.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(ii) Explain why sulfur does **not** follow this trend.

RocketRevise

(2)

.....

.....

.....

.....

.....

(Total for question = 10 marks)

www.RocketRevise.com

www.RocketRevise.com

Q75.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which diagram represents the electronic structure of a nitrogen atom?

		1s	2s	2p		
☐	**A**	↑↓	↑	↑↓	↑	↑
☐	**B**	↑↓	↑	↑↓	↑↓	
☐	**C**	↑↓	↑↓	↑	↑	↑
☐	**D**	↑↓	↑↓	↑↓	↑	

RocketRevise

(Total for question = 1 mark)

Q76.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which equation represents the **second** ionisation energy of magnesium?

- ☐ **A** $Mg(g) \rightarrow Mg^{2+}(g) + 2e^-$
- ☐ **B** $Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$
- ☐ **C** $Mg^+(s) \rightarrow Mg^{2+}(s) + e^-$
- ☐ **D** $Mg(g) \rightarrow Mg^+(g) + e^-$

(Total for question = 1 mark)

RocketRevise

Q77.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The first four ionisation energies of the elements gallium (Ga), indium (In), germanium (Ge) and tin (Sn) are shown.

Which values are the first four ionisation energies of gallium?

- ☐ **A** 557 1821 2705 5200
- ☐ **B** 579 1979 2963 6200
- ☐ **C** 709 1412 2943 3930

www.RocketRevise.com

www.RocketRevise.com

☒ **D** 762 1537 3302 4411

(Total for question = 1 mark)

Q78.

Samples of germanium consist of five isotopes, four of which are stable.
The fifth isotope, germanium-76, is radioactive but with a very long half-life.

This means that the isotopic composition is almost constant.
The table shows data for four of the isotopes.

RocketRevise

Atom	^{70}Ge	^{73}Ge	^{74}Ge	^{76}Ge
Mass number	70	73	74	76
Abundance / %	20.5	7.8	36.5	7.8

(a) (i) Calculate the mass number of the fifth isotope of germanium.

The relative atomic mass of germanium is 72.6.
You must show your working.

(3)

RocketRevise

(ii) Give the main reason why the answer in (a)(i) should not be quoted to more than 2 significant figures.

(1)

.....

.....

.....

(b) The radioactive decay of germanium-76 forms stable atoms of selenium-76.

Compare and contrast the atomic structure of an atom of germanium-76 with an atom of selenium-76.

(4)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 8 marks)

Q79.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Sulfur has a higher melting temperature than white phosphorus, P_4.

Which is a possible reason for this?

- ☒ **A** sulfur molecules have a greater permanent dipole than phosphorus molecules
- ☒ **B** sulfur has stronger covalent bonds than phosphorus
- ☒ **C** sulfur has a giant covalent structure and phosphorus has a simple molecular structure
- ☒ **D** sulfur molecules have more electrons than phosphorus molecules

(Total for question = 1 mark)

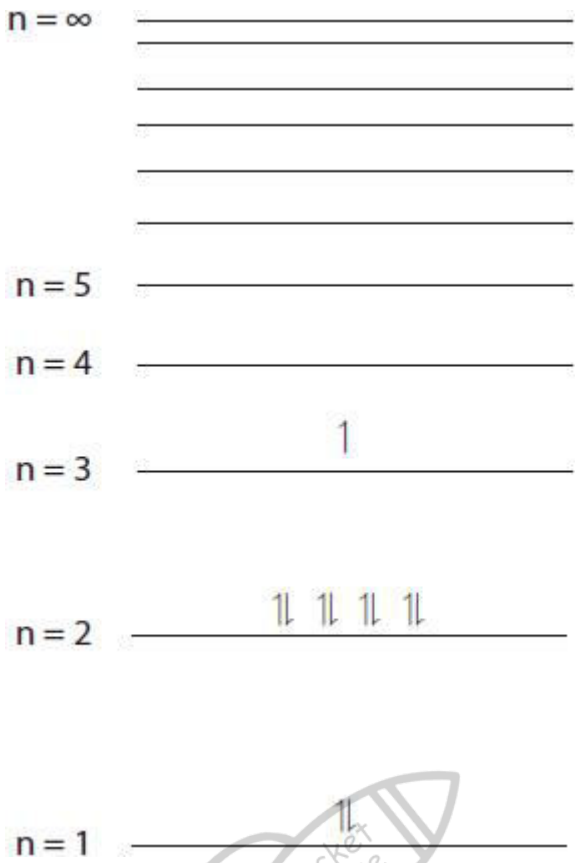
RocketRevise

Q80.

The diagram shows the arrangement of electrons in the energy levels of a sodium atom.

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

(a) Give the reason why the two electrons in the n = 1 energy level are represented by arrows pointing in opposite directions.

(1)

.....

.....

.....

.....

(b) Explain whether or not all of the electrons in the n = 2 energy level have the same energy.

(2)

.....

.....

.....

.....

.....

.....

(c) Name the shape of the orbital occupied by the electron in the n = 3 energy level.

(1)

.....

www.RocketRevise.com

www.RocketRevise.com

(Total for question = 4 marks)

Q81.

This question is about ionisation energy.

(a) Write the equation for the third ionisation energy of calcium.

Include state symbols.

(3)

RocketRevise

(b) (i) The following data show the first seven ionisation energies for an element in Period 3. Deduce, with a reason, the identity of the element.

(2)

Ionisation number	1	2	3	4	5	6	7
Energy / kJ mol^{-1}	1000	2251	3361	4564	7012	8496	27107

.....

.....

.....

.....

(ii) State the general trend in first ionisation energies across Period 3.

(1)

.....

.....

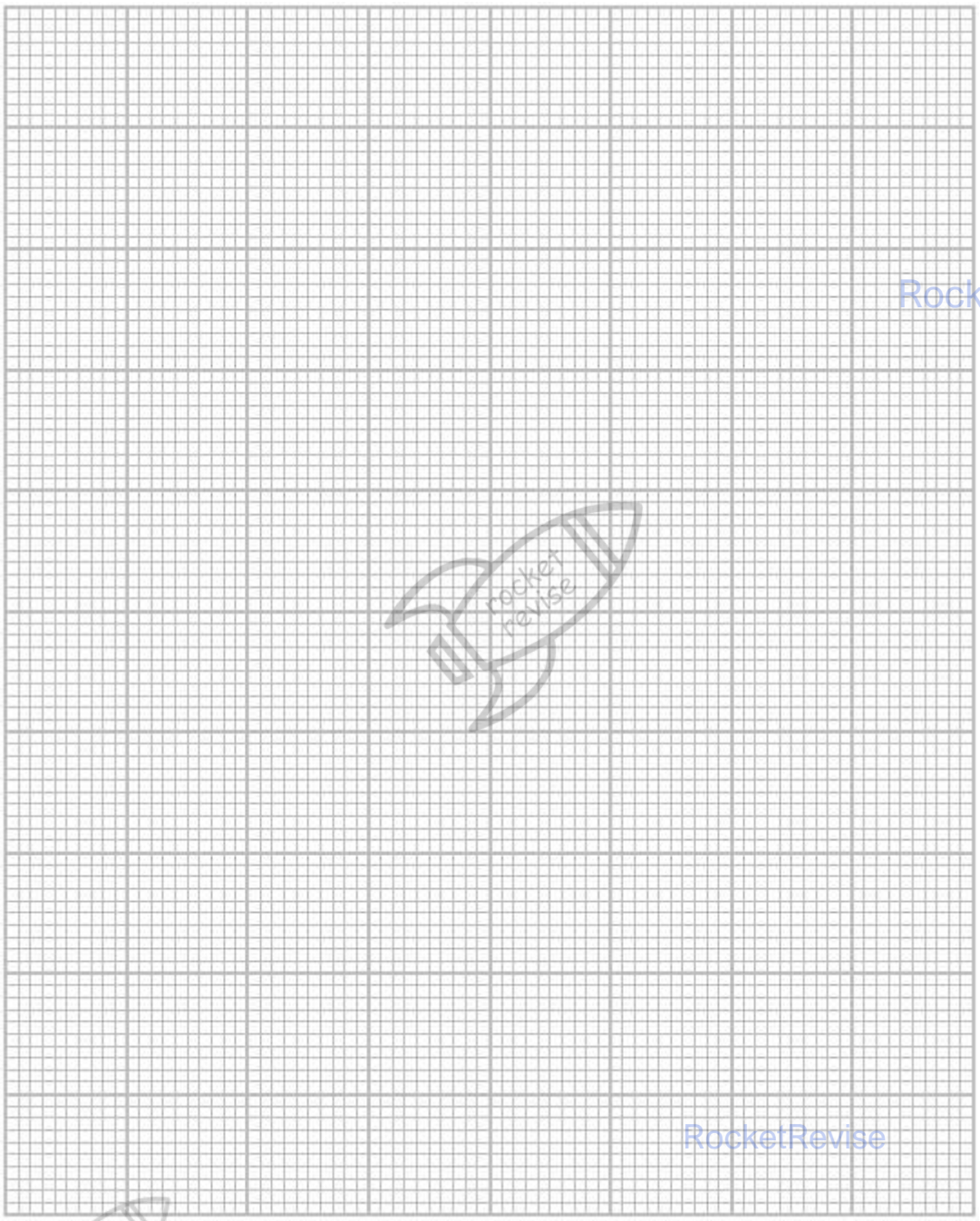
(c) (i) Plot the following first ionisation energies of the elements in Period 2 on the grid below.

(3)

www.RocketRevise.com

www.RocketRevise.com

Element	Li	Be	B	C	N	O	F	Ne
1st ionisation energy / kJ mol^{-1}	520	900	801	1086	1402	1314	1681	2081



(ii) Explain the difference in first ionisation energies between beryllium and boron.

(2)

.....

.....

.....

.....

(iii) Explain the difference in first ionisation energies between carbon and nitrogen.

www.RocketRevise.com

www.RocketRevise.com

(2)

.....

.....

.....

.....

(Total for question = 13 marks)

RocketRevise

Q82.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which row shows the numbers of neutrons and electrons in a bromide ion $^{79}Br^-$?

		Number of neutrons	Number of electrons
☐	A	44	35
☐	B	44	36
☐	C	46	35
☐	D	46	36

(Total for question = 1 mark)

Q83.

RocketRevise

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the correct order for the processes in a mass spectrometer?

☐	A	deflection	detection	ionisation	vaporisation
☐	B	vaporisation	ionisation	deflection	detection
☐	C	ionisation	vaporisation	deflection	detection
☐	D	vaporisation	deflection	ionisation	detection

www.RocketRevise.com

www.RocketRevise.com

(Total for question = 1 mark)

RocketRevise

RocketRevise

www.RocketRevise.com